

# Shadows of the Swampland:

## Dark Matter as Geometric Surface Tension and Dark Energy as Vacuum Relaxation

Aaron M. Schutz  
*Independent Researcher*

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### Abstract

We complete the Axiomatic Physical Homeostasis (APH) cosmology by analytically deriving the *Dark Sector* of the universe from first principles of Vacuum Material Science. We demonstrate that Dark Matter and Dark Energy are not independent particle or field theoretic entities, but conjugate geometric effects arising from the Sedenion-to-Octonion phase filtration. We identify Dark Matter as the *Surface Tension* of the  $G_2$  domain walls separating the Octonionic vacuum filaments from the excluded Sedenion voids. By integrating the surface virial term into the galactic dynamic equations, we derive the flat rotation curves of spiral galaxies without invoking particulate halos. Concurrently, we identify Dark Energy as the *Elastic Relaxation* of the vacuum stiffness from the primordial limit  $\Gamma = \pi$  to the stable Octonionic value  $\Gamma \approx 2.91$ . We derive a time-dependent equation of state parameter  $w(z)$  for the vacuum, predicting a specific deviation from  $\Lambda$ CDM that resolves the Hubble Tension.

## 1 Introduction: The Material Science of the Vacuum

Standard Cosmology ( $\Lambda$ CDM) parameterizes the unknown components of the universe as a collisionless fluid (Dark Matter) and a constant vacuum energy density (Dark Energy). Despite their success in fitting the Cosmic Microwave Background (CMB) power spectrum, these parameters lack a fundamental microphysical derivation.

The APH framework replaces these effective parameters with the mechanical properties of the vacuum geometry. We posit that the universe is a *composite material* resulting from the topological filtration of the Sedenion algebra  $\mathbb{S}$ . This composite structure consists of:

1. **The Bulk (Voids):** Regions of excluded Sedenion zero divisors ( $\mathcal{Z}(\mathbb{S})$ ), characterized by high stiffness  $\Gamma_{\mathcal{S}} = \pi$ .
2. **The Filament (Matter):** Regions of stable Octonionic geometry ( $\mathbb{O}$ ), characterized by relaxed stiffness  $\Gamma_{\mathbb{O}} \approx 2.91$ .
3. **The Interface (Halo):** The domain walls separating these phases, possessing a macroscopic  $G_2$  topology.

In this paper, we derive the gravitational anomalies attributed to the Dark Sector as the manifestation of *Geometric Surface Tension* (Dark Matter) and *Stiffness Relaxation* (Dark Energy).

## 2 Dark Matter: The Surface Tension of $G_2$ Domain Walls

We propose that galactic halos are not clouds of unseen particles, but the elastic potential wells created by the  $G_2$  domain walls wrapping the visible matter.

## 2.1 The Geometric Surface Virial Theorem

Consider a galaxy as a virialized system of visible mass  $M_{bar}$  enclosed within a geometric boundary  $\partial V$  (the  $G_2$  interface). The standard Virial Theorem  $2T + W = 0$  is modified by the surface pressure term exerted by the Sedenion void. The generalized tensor virial equation is:

$$\frac{1}{2} \frac{d^2 I_{jk}}{dt^2} = 2T_{jk} + W_{jk} + S_{jk} \quad (1)$$

where  $S_{jk}$  is the surface stress tensor given by:

$$S_{jk} = \oint_{\partial V} x_j (P_{void} \delta_{km} - \sigma_{wall} n_k n_m) dS_m \quad (2)$$

Here,  $P_{void}$  is the pressure of the Sedenion bulk and  $\sigma_{wall}$  is the surface tension of the  $G_2$  domain wall.

## 2.2 Derivation of Flat Rotation Curves

We assume the galaxy is a disk embedded in a spherical  $G_2$  interface. The Sedenion void exerts an inward *Geometric Pressure* due to the stiffness gradient  $\nabla \Gamma$  between the void ( $\pi$ ) and the filament (2.91). This pressure gradient creates an effective confining force  $\mathbf{F}_{geom}$  acting on the visible matter. For a spherically symmetric domain wall, the surface tension contribution to the gravitational potential  $\Phi$  scales logarithmically with radius due to the codimension-1 nature of the defect (analogous to the potential of a cosmic string or domain sheet).

$$\Phi_{total}(r) = \Phi_{Newton}(r) + \Phi_{geom}(r) = -\frac{GM_{bar}}{r} + v_0^2 \ln \left( \frac{r}{r_s} \right) \quad (3)$$

where  $v_0$  is a characteristic velocity determined by the domain wall tension  $\sigma_{wall}$ . The orbital velocity  $v(r)$  is given by the equilibrium condition  $v^2/r = d\Phi/dr$ :

$$\frac{v(r)^2}{r} = \frac{GM_{bar}}{r^2} + \frac{v_0^2}{r} \quad (4)$$

$$v(r) = \sqrt{\frac{GM_{bar}}{r} + v_0^2} \quad (5)$$

**Prediction:** As  $r \rightarrow \infty$ , the Newtonian term vanishes, and the rotation curve asymptotically flattens to  $v(r) \rightarrow v_0$ . Unlike the Tully-Fisher relation which is empirical, APH derives  $v_0$  from the fundamental tension of the  $G_2$  manifold:

$$v_0^2 \propto G \cdot \sigma_{wall} \cdot R_{scale} \quad (6)$$

Thus, *Dark Matter* is simply the elastic energy stored in the boundary of the cosmic void.

## 2.3 Topology of the Dark Sector

Since the domain wall is isometric to  $G_2$  and fibers over the Stiefel manifold  $V_2(\mathbb{R}^7)$ , we predict that the large-scale distribution of the gravitational potential (the *Dark Matter Halo*) must exhibit the topological invariants of  $V_2(\mathbb{R}^7)$ . Specifically, the mass distribution is not Gaussian but follows the homology of the zero-divisor manifold:

$$\rho_{DM}(\mathbf{x}) \sim \text{Characteristic Class}(V_2(\mathbb{R}^7)) \quad (7)$$

This predicts specific non-Gaussianities in weak lensing maps that correspond to the Betti numbers of the Stiefel manifold.

### 3 Dark Energy: The Relaxation of Vacuum Stiffness

We re-interpret Cosmic Acceleration not as the effect of a constant  $\Lambda$ , but as the mechanical relaxation of the vacuum stiffness  $\Gamma(t)$ .

#### 3.1 The Time-Dependent Polytopic Index

The universe emerged from the Big Bang in the Sedenion Phase with maximum geometric stiffness  $\Gamma_S = \pi$ . It is relaxing toward the stable Octonionic Phase  $\Gamma_{\mathbb{O}} \approx 2.91$ . We model this relaxation as an exponential decay in cosmic time  $t$ :

$$\Gamma(t) = \Gamma_{\mathbb{O}} + (\Gamma_S - \Gamma_{\mathbb{O}})e^{-t/\tau} \quad (8)$$

where  $\tau$  is the characteristic relaxation timescale of the geometry.

#### 3.2 The Vacuum Equation of State $w(z)$

The pressure  $P$  and density  $\rho$  of the vacuum fluid are related by the instantaneous stiffness:

$$P_{vac} = -\rho_{vac}c^2 + P_{elastic} \quad (9)$$

The elastic rebound pressure arises from the deviation of stiffness from the stability floor. Using the Friedmann acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P/c^2) \quad (10)$$

We derive the effective Equation of State parameter  $w = P/\rho c^2$ . In APH, this is not constant (-1), but evolves with redshift  $z$ :

$$w(z) = -1 + \Delta w_0 \left( \frac{\Gamma(z) - \Gamma_{\mathbb{O}}}{\Gamma_S} \right) \quad (11)$$

Since  $\Gamma(z)$  is decaying from  $\pi$  to 2.91, the term in the parentheses is positive but decreasing. **Prediction:** The vacuum behaves as a “quintessence” field with  $w(z) > -1$  (approaching -1 from above). This implies there is no *Phantom Crossing* ( $w < -1$ ) and no *Big Rip*. The acceleration is the transient unloading of the *Big Bang Spring*.

#### 3.3 Resolution of the Hubble Tension

The tension between early-universe ( $H_0$  from CMB) and late-universe ( $H_0$  from Supernovae) measurements arises from assuming a constant  $\Lambda$ . The APH model introduces a stiff energy component in the early universe ( $\Gamma \approx \pi$ ) that decays into the late universe.

$$H^2(z) = H_0^2 \left[ \Omega_m(1+z)^3 + \Omega_{vac} \exp \left( \int_0^z \frac{3(1+w(z'))}{1+z'} dz' \right) \right] \quad (12)$$

By fitting the relaxation time  $\tau$ , APH naturally accommodates a higher  $H_0$  in the local (relaxed) universe compared to the early (stiff) universe, resolving the discrepancy without new physics.

### 4 Conclusion

We have unified the Dark Sector under the geometry of the Sedenion-Octonion phase transition.

- **Dark Matter** is the surface tension of the  $G_2$  domain walls ( $V_2(\mathbb{R}^7)$  topology), providing the extra centripetal force for galactic rotation.

- **Dark Energy** is the relaxation of vacuum stiffness from  $\pi$  to 2.91, driving cosmic acceleration via elastic rebound.

This completes the APH tetralogy, offering a purely geometric, finite, and consistent description of the cosmos from the Big Bang to the present epoch.

## References

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